M378K: April 13th, 2026.

More About Maximum Likelihood Estimation.

Def'n. An estimator $\hat{\theta} = \hat{\theta}(y_1, \dots, y_n)$ is called the **maximum likelihood estimator (MLE)** if it satisfies that for any $\hat{\theta}' = \hat{\theta}'(y_1, \dots, y_n)$ we have

$$L(\hat{\theta}; y_1, \dots, y_n) \geq L(\hat{\theta}'; y_1, \dots, y_n)$$

for all $y_1, \dots, y_n$

Example. Normal.

$$Y_1, Y_2, \dots, Y_n \sim N(\mu, \sigma)$$

the pdf: $f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$ for all $y \in \mathbb{R}$

$$L(\mu, \sigma; y_1, \dots, y_n) = \prod_{i=1}^{n} \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y_i-\mu}{\sigma}\right)^2} \right)$$

unknown

$$= (2\pi)^{-\frac{n}{2}} \cdot \sigma^{-n} \cdot e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2}$$

$$\propto e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2}$$

$$\ell(\mu; y_1, \dots, y_n) = \text{const} - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2$$

$$\ell'(\mu; y_1, \dots, y_n) = +\frac{1}{\cancel{2}\sigma^2}\sum_{i=1}^{n}\cancel{2}(y_i-\mu)(+1) = 0$$

$$\sum_{i=1}^{n}(y_i-\mu) = 0$$

$$\sum_{i=1}^{n} y_i - n\mu = 0$$

$$\boxed{\hat{\mu}_{MLE} = \bar{Y}}$$

$\square$

Example. $Y_1, Y_2, \dots, Y_n \sim \underline{U(0,\theta)}$ $\boxed{\theta > 0 \text{ unknown}}$

The pdf of $U(0,\theta)$:

$$f^{\theta}(y) = \frac{1}{\theta}\mathbb{1}_{[0,\theta]}(y) = \boxed{\frac{1}{\theta}\mathbb{1}_{\{0 \le y \le \theta\}}}$$

$$L(\theta; y_1, \dots, y_n) = \prod_{i=1}^{n}\left(\frac{1}{\theta}\mathbb{1}_{\{0 \le y_i \le \theta\}}\right)$$

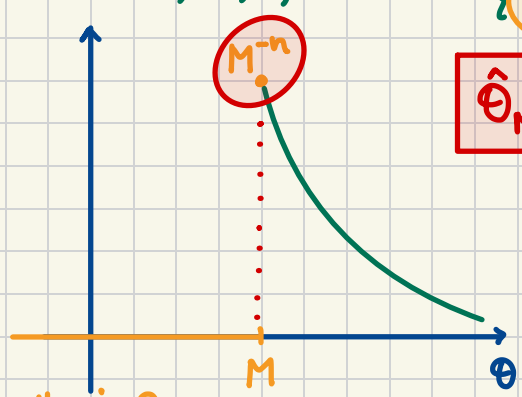
$$= \left(\frac{1}{\theta}\right)^n \cdot \mathbb{1}_{\{0 \le y_1, \dots, y_n \le \theta\}}$$

$$= \left(\frac{1}{\theta}\right)^n \boxed{\mathbb{1}_{\{0 \le \min(y_1, \dots, y_n)\}}} \cdot \mathbb{1}_{\{\max(y_1, \dots, y_n) \le \theta\}}$$

"assume $y_1, \dots, y_n \ge 0$"

$$= \theta^{-n} \cdot \mathbb{1}_{\{\max(y_1, \dots, y_n) \le \theta\}}$$

$$L(\theta; y_1, \dots, y_n) = \theta^{-n} \cdot \mathbb{1}_{\{\overset{M}{y_{(n)}} \le \theta\}}$$



the $\mathbb{1}$ is 0

$$\boxed{\hat{\theta}_{MLE} = \max(Y_1, \dots, Y_n) = Y_{(n)}}$$

Q: Is it unbiased?

⋮

Note: Moment Matching.

theoretical mean = sample mean

$$\frac{\theta}{2} = \bar{Y} \implies \boxed{\hat{\theta}_{MM} = 2\bar{Y}}$$

Example. Let $Y_1, \dots, Y_n$ be a random sample from a dist'n w/ density

$$f^{\theta}(y) = \boxed{\theta \cdot y^{\theta-1} \cdot \mathbb{1}_{[0,1]}(y)}$$

for some unknown positive parameter θ.

Find the MLE for θ.

$\longrightarrow$: $$L(\theta; y_1, \dots, y_n) = \prod_{i=1}^{n} (\theta \cdot y_i^{\theta-1}) = \theta^n \left(\prod_{i=1}^{n} y_i\right)^{\theta-1}$$

assuming $y_1, \dots, y_n \in [0,1]$

$$\ell(\theta; y_1, \dots, y_n) = n \cdot \ln(\theta) + (\theta - 1) \cdot \sum_{i=1}^{n} \ln(y_i)$$

$$\ell'(\theta; y_1, \dots, y_n) = n \cdot \frac{1}{\theta} + \sum_{i=1}^{n} \ln(y_i) = 0$$

$$\frac{n}{\theta} = -\sum_{i=1}^{n} \ln(y_i)$$

$$\boxed{\hat{\theta}_{MLE} = \frac{-n}{\sum_{i=1}^{n} \ln(y_i)}}$$

$\square$

Example. $\boxed{\text{CAS Exam 3, Spring 2007.}}$

Consider a random sample $Y_1, \dots, Y_n$ from a dist'n w/ the pdf

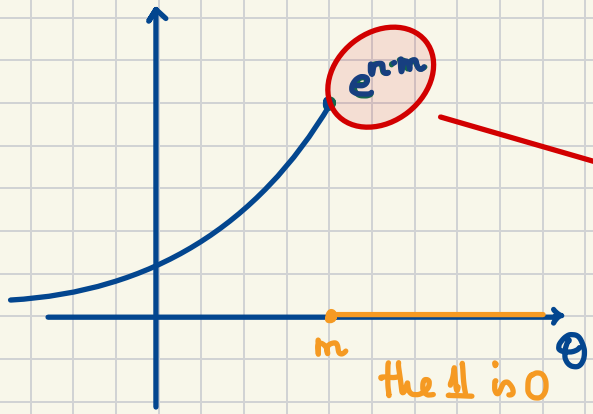
$$f^{\theta}(y) = \boxed{e^{-y+\theta} \cdot \mathbb{1}_{(\theta, +\infty)}(y)}.$$

Find the MLE for θ.

$$\longrightarrow: \; L(\theta; y_1, \dots, y_n) = \prod_{i=1}^{n} \left(e^{-y_i + \theta} \cdot \mathbb{1}_{(\theta, +\infty)}(y_i) \right)$$

$$= e^{-\sum_{i=1}^{n} y_i} \cdot \boxed{e^{n\theta} \cdot \mathbb{1}_{\{\theta < \underbrace{\min(y_1, \dots, y_n)}_{y_{(1)}}\}}}$$

$y_{(1)}$ m



$$\boxed{\hat{\theta}_{MLE} = Y_{(1)}}$$

$\square$